

ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.*(An ISO 9001:2015 Certified College)***DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING****B. TECH DEGREE FIRST INTERNAL TEST-NOVEMBER 2022**

Fifth Semester

CST 305 SYSTEM SOFTWARE

Time: 2 hours

Max. Marks: 50

PART A

Answer all questions (3 x 5 = 15 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
1	Distinguish between an assembler and a compiler. Which are the different types of compilers?	3	K2	CO1 & K2
2	Differentiate between the Linker and Loader.	3	K2	CO1 & K2
3	What are the uses of OPTAB and SYMTAB during the assembling process? Specify the uses of each during pass 1 and pass2 of a two pass assembler.	3	K2	CO3 & K3
4	What are assembler directives? List out any five assembler directives in SIC.	3	K1	CO2 & K3
5	Write a sequence of instructions for SIC to set $\text{ALPHA} = \text{BETA} * 9 + \text{GAMMA}$	3	K3	CO2 & K3

PART B

Answer five questions (7 x 5 = 35 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL																
Answer any two questions																				
6	Explain the architecture of a SIC machine.	7	K1	CO2 & K3																
7	Translate (by hand) the following assembly program to <u>SIC object code</u>. Starting address program is 1000(H) . The output format will contains H record, T record, and E record. LDA -00, ADD - 18, STA -0C, SUB - 1C, MUL - 20 <table><tr><td>1000</td><td>LDA</td><td>ALPHA</td><td>000015</td></tr><tr><td>1003</td><td>ADD</td><td>BETA</td><td>18018</td></tr><tr><td>1006</td><td>STA</td><td>TEMP</td><td>0C1021</td></tr><tr><td>1009</td><td>LDA</td><td>GAMMA</td><td>0D101B</td></tr></table>	1000	LDA	ALPHA	000015	1003	ADD	BETA	18018	1006	STA	TEMP	0C1021	1009	LDA	GAMMA	0D101B	7	K3	CO2 & K3
1000	LDA	ALPHA	000015																	
1003	ADD	BETA	18018																	
1006	STA	TEMP	0C1021																	
1009	LDA	GAMMA	0D101B																	

AC $\alpha + \beta \rightarrow \text{temp}$
 $(\gamma - \delta) * \text{temp} \rightarrow M$

1000		SUB	DELTA	10101E		
1004		MUL	TEMP	201021		
1018		STA	M	001024		
1015	ALPHA	WORD +3	4	000100		
1018	BETA	WORD +3	6	000110		
1018	GAMMA	WORD +3	7	000111		
101E	DELTA	WORD +3	5	000101		
1021	TEMP	RESW +3	1			
1024	M	RESW +3	1			
1025		END				
8	Write SIC instructions to swap the values of ALPHA and BETA.			7	K3	CO2 & K3
Answer any two questions						
9	What are the various addressing modes supported by SIC/XE? With the help of an example, explain how to find target address during assembling in each case.			7	K2	CO3 & K3
10	Translate (by hand) the following assembly program to SIC/XE object code. Starting address program is 1000(H) .Also assume Opcode for instruction. The output format will contains H record, T record, and E record 1000 1000 1003 96 96 96 12 X STRCP2 FIRST MOVECH STR1 STR2 START LDT LDX LDCH STCH TIXR JLT BYTE RESB END 1000 #11 #0 STR1,X STR2,X T MOVECH C'TEST STRING' 11 FIRST label opcode operands			7	K3	CO2 & K3
11	Assemble the following instruction indicating the instruction formats used: a) RMO S,A b) +JSUB RDREC c) LDA #1 Assume that the value of RDREC is 2000.			7	K3	CO3 & K3
		Opcode	Machine code			
		RMO	AC			

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		JSUB	48			
		LDA	10			
		Register				
		A	0			
		S	4			
Mandatory Question						
12	With the aid of an algorithm explain the Second pass of a Two Pass Assembler			7	K2	CO3 & K3



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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B. TECH DEGREE SECOND INTERNAL TEST- DECEMBER 2022

Fifth Semester

CST 305 SYSTEM SOFTWARE

Time: 2 hours

Max. Marks: 50

PART A

Answer all questions (3 x 5 = 15 Marks)

Qn. No.	Question	Marks	K level	CO & KL
1 ✓	Give the structure and purpose of Modification record and Define record.	3	K1	CO3 & K2
2 ✓	What are the uses of the assembler directives EXTDEF and EXTREF?	3	K1	CO3& K2
3 ✓	In system software, what is LTORG and how does it work?	3	K1	CO3& K2
4 ✓	How does an assembler handle literals?	3	K2	CO4 & K2
5 ✓	What is the need for a multi-pass assembler?	3	K2	CO3 & K2

PART B

Answer five questions (7 x 5 = 35 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
Answer any two questions				
6	Define <u>Program Blocks</u> . Explain the different program blocks within an assembly program using an example and diagram. Also mention how the object code program for a program block is organized.	7	K2	CO4 & K2
7 ✓	Explain in detail about the <u>Symbol-Defining Statements</u>	7	K1	CO4 & K2
8	What are <u>control sections</u> ? Illustrate with an example, how control sections are used and linked in an assembly language program.	7	K2	CO4 & K2
Answer any two questions				
9 ✓	Define <u>Absolute loader</u> and outline the algorithm for it.	7	K1	CO5 & K2
10	What is <u>ESTAB</u> in linker loader? Explain the working of ESTAB with respect to the linker loader algorithm.	7	K2	CO5 & K2
11 ✓	Write the algorithm for the <u>pass two</u> of a <u>two pass linker loader</u> .	7	K1	CO5 & K2
Mandatory Question				
12	What is the use of <u>Modification record</u> and <u>Relocation bit</u> in Relocating loader? Explain with an example.	7	K2	CO3 & K2

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Regular and Supplementary Examination December 2022 (2019 Scheme)

Course Code: CST 305**Course Name: SYSTEM SOFTWARE**

Max Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

- | | | |
|----|--|---|
| 1/ | Distinguish between interpreter and compiler. | 3 |
| 2/ | List any three registers available in SIC machine along with their purpose | 3 |
| 3/ | List out the basic functions of Assemblers | 3 |
| 4/ | Write an SIC program to swap the values of ALPHA and BETA | 3 |
| 5/ | With an example explain any two symbol defining statements? | 3 |
| 6/ | Define a program block. How it is created? | 3 |
| 7/ | List the basic functions of a loader | 3 |
| 8/ | Define a modification record along with its structure | 3 |
| 9 | Illustrate the concept of macro definition with an example | 3 |
| 10 | What are the two parts of a device driver? | 3 |

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

- | | | |
|-------|--|---|
| 11 a/ | Write notes on SIC machine architecture | 8 |
| b/ | What are assembler directives? List any four assembler directives in SIC machine. | 6 |
| 12 a) | Elucidate the architecture of SIC/XE machine | 8 |
| b) | Compare the features of Standard SIC and SIC/XE architecture | 6 |

Module -2

- | | | |
|-------|---|---|
| 13 a/ | Write a sequence of instructions for SIC/XE to divide BETA by GAMMA , setting ALPHA to the integer portion of the quotient and DELTA to the remainder. Use register to-register instructions to make the calculation as efficient as possible. | 5 |
|-------|---|---|

- b) Illustrate the use and structure of three records used in object program 3
- c) Explain the data structures used and their purposes in a two-pass assembler 6
- 14 a) Suppose ALPHA is an array of 100 words. Write SIC/XE program to set all array elements to zero. 6
- b) Design an algorithm for pass 1 operations of a two pass assembler for SIC architecture. 8

Module -3

- 15 a) Write short notes on MASM assembler 7
- b) Employ the following code to explain the concept of multipass assembler 7
- | | | | |
|---|---|------|------|
| 1 | A | EQU | B/2 |
| 2 | B | EQU | C-D |
| 3 | E | EQU | D-1 |
| 4 | D | RESB | 4096 |
| 5 | C | EQU | * |
- 16 a) Outline in detail **Load-and-go** Single Pass Assembler Algorithm 7
- b) What are control sections? Illustrate with an example, how control sections are used and linked in an assembly language program. 7

Module -4

- 17 a) Write notes on machine independent loader features. ✓ 8
- b) Which are the data structures used during the operation of a linking loader? Write the algorithm for Pass 2 of a Linking Loader 6
- 18 a) Give the algorithm for an absolute loader 6
- b) Write notes on the different loader design options 8

Module -5

- 19 a) Explain the working of One pass Macro Processor along with algorithm 6
- b) Explain the types of macro with example 8
- 20 a) Write notes on text editor 7
- b) Discuss the features of device drivers 7
